ICE-4107: Microwave Engineering

Lecture 6

Plane-wave propagation in free space and lossless dielectric

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Plane-Wave Propagation in Free Space

- The electromagnetic wave being propagated in free space near the surface of the earth is divided into two parts: the ground wave and the sky wave or ionosphere wave.
- The ground wave is further divided into a direct wave, an earth-reflected wave, and a surface wave.

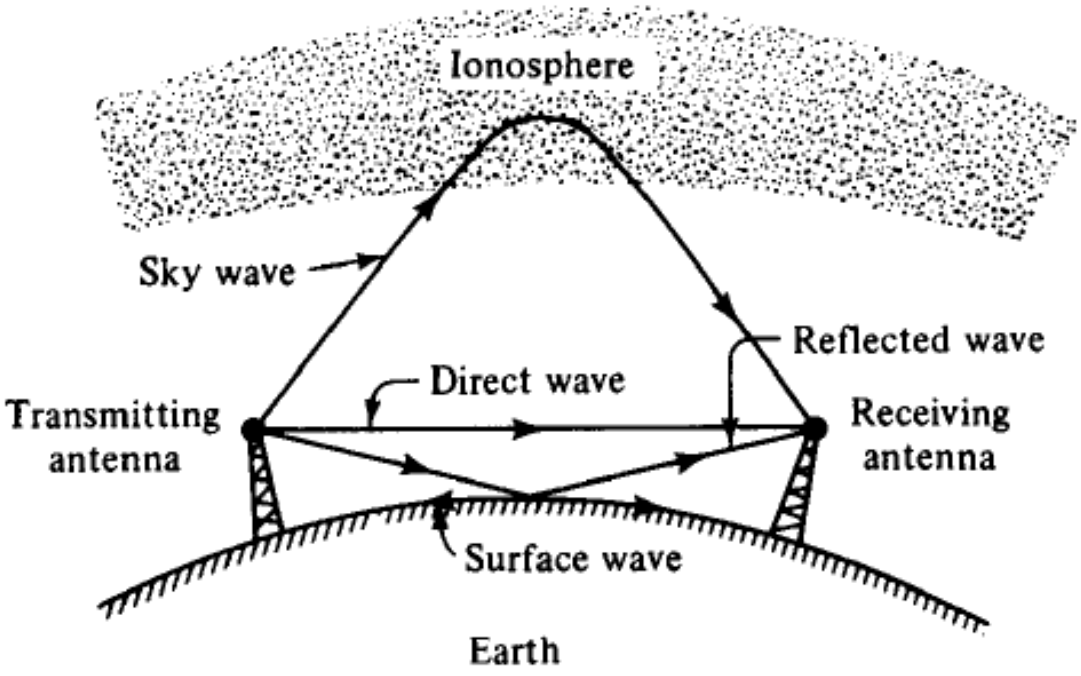


Figure 2-4-1 Wave components near the surface of the earth.

Plane-Wave Propagation in Free Space

- Figure 2-4-1 shows the wave components of electromagnetic wave from a nondirectional antenna to a receiving station.
- The ionosphere is that region of the earth's atmosphere in which the constituent gases are ionized by solar radiation. This region extends from about 50 km above the earth to several earth radii and has different layers designated as C, D, E, and F layers in order of height.
- The electron-density distribution of each layer varies with the time of day, season, year, and the geographical location.

Plane-Wave Propagation in Free Space

- The electron density determines the reflection and refraction of microwaves. For vertical incidence, the critical frequency is given by $F_{cr} = 9\sqrt{N_{\max}}\ \ Hz$
- This means that a microwave of frequency F_{cr} will be reflected back to the earth if the electron density is equal to or higher than the required maximum electron density N_{max} (electrons per cubic meter).
- The sky wave reaches the receiving station after reflection from the ionosphere.

Plane-Wave Propagation in Free Space

- Although important in many communication systems, the sky wave need not be considered in most microwave applications because a wavelength shorter than about 4 m (74.9 MHz) will not return to the earth from the ionosphere.
- The wave is reflected from the earth in such a way as to reach the receiver. Energy radiated from the nondirectional antenna of Fig. 2-4-1 strikes the earth at all points between the base of the antenna and the horizon (sphere).
- The surface wave is a wave diffracted around the surface of the earth or guided by the ground-air interface.

Plane-Wave Propagation in Free Space

- This component is important at broadcast frequencies; at microwave frequencies, however, the surface wave is rapidly attenuated, and at a distance of 2 km from the antenna it has an amplitude of only a fraction of 1% of the direct wave. This component must be considered in blind-landing systems in which ranges of less than 2 km are important.
- The direct wave travels a nearly straight path from the transmitting antenna to the receiving station.

Plane-Wave Propagation in Free Space

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Plane-Wave Propagation in Free Space

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Plane-Wave Propagation in Free Space

- When power radiates from the transmitting antenna, the power density carried by the spherical wave will decrease with distance as the energy in the wave spreads out over an ever-increasing surface area as the wave progresses. The power density is given by

$$p_d = \frac{p_t g_t}{4\pi R^2} \quad \text{W/m}^2 \qquad (2\text{-}4\text{-}2)$$

where p_t = transmitting power in watts
g_t = transmitting antenna gain (numerical)
R = distance between antenna and field point in meters

The power received by the receiving antenna is given as

$$p_r = p_d A_e = \left(\frac{p_t g_t}{4\pi R^2}\right)\left(\frac{\lambda_2}{4\pi} g_r\right) \quad \text{watts} \qquad (2\text{-}4\text{-}3)$$

where $A_e = \frac{\lambda^2}{4\pi} g_r$ = effective antenna aperture in square meters

$\frac{\lambda^2}{4\pi} = A_a$ = isotropic antenna aperture in square meters

g_r = receiving antenna gain (numerical)

Plane-Wave Propagation in Free Space

Figure 2-4-2 shows the relationships of electromagnetic energy transmission in free space between two antennas.

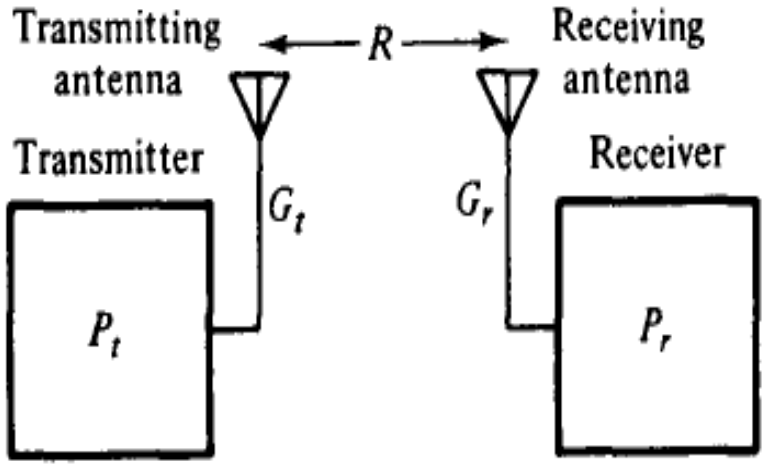


Figure 2-4-2 Electromagnetic energy transmission between two antennas.

If the received power is expressed in terms of decibels, Eq. (2-4-3) becomes

$$P_r = P_t + G_t + G_r - 20 \log \left(\frac{4\pi R}{\lambda}\right) \qquad \text{dBW} \qquad (2\text{-}4\text{-}4)$$

where P_t is in dBW, G_t and G_r are in decibels (dB). The term $20 \log (4\pi R/\lambda)$ is well known as the free-space attenuation in decibels.

Plane-Wave Propagation in Lossless Dielectric

The lossless dielectric, which is often called the good or perfect dielectric, is characterized by $\sigma = 0$. Hence the intrinsic impedance for a lossless dielectric can be expressed in terms of air. This is

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_r \epsilon_0}} = \frac{377}{\sqrt{\epsilon_r}} \quad \text{ohms} \tag{2-4-5}$$

The attenuation constant α is zero, and the phase constant β is given by

$$\beta = \omega\sqrt{\mu\epsilon} \tag{2-4-6}$$

The phase velocity is expressed by

$$v_p = \frac{1}{\sqrt{\mu\epsilon}} \tag{2-4-7}$$

Plane-Wave Propagation in Lossless Dielectric

- The lossy media are characterized by $\sigma \neq 0$. There are three types of lossy media: good conductor, poor conductor, and lossy dielectric. The presence of a loss in the medium introduces wave dispersion by conductivity.

The electric and magnetic wave equations in the frequency domain as shown in Eqs. (2-1-20) and (2-1-21) are repeated here:

$$\nabla^2 \mathbf{E} = j\omega\mu(\sigma + j\omega\epsilon)\mathbf{E} \qquad (2\text{-}5\text{-}1)$$

$$\nabla^2 \mathbf{H} = j\omega\mu(\sigma + j\omega\epsilon)\mathbf{H} \qquad (2\text{-}5\text{-}2)$$

For one dimension in the positive z direction, they become

$$\frac{\partial^2 E_x}{\partial z^2} = j\omega\mu(\sigma + j\omega\epsilon)E_x \qquad (2\text{-}5\text{-}3a)$$

Plane-Wave Propagation in Lossless Dielectric

$$\frac{\partial^2 H_y}{\partial z^2} = j\omega\mu(\sigma + j\omega\epsilon)H_y \qquad (2\text{-}5\text{-}3b)$$

The complex-frequency solutions would be given by

$$E_x = E_0 e^{-\alpha z}\cos(\omega t - \beta z) \qquad (2\text{-}5\text{-}4)$$

$$H_y = \frac{E_0}{\eta} e^{-\alpha z}\cos(\omega t - \beta z) \qquad (2\text{-}5\text{-}5)$$

where $\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = \alpha + j\beta$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$$

Plane-Wave in Good Conductor

• A good conductor is defined as one having a very high conductivity; consequently, the conduction current is much larger than the displacement current.

• The energy transmitted by the wave traveling through the medium will decrease continuously as the wave propagates because ohmic losses are present. Expressed mathematically, a good conductor requires the criterion

$$\sigma \gg \omega\epsilon \tag{2-5-6}$$

The propagation constant γ is expressed as

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = j\omega\sqrt{\mu\epsilon}\sqrt{1 - j\frac{\sigma}{\omega\epsilon}}$$

$$= j\omega\sqrt{\mu\epsilon}\sqrt{-j\frac{\sigma}{\omega\epsilon}} \qquad \text{for } \frac{\sigma}{\omega\epsilon} \gg 1$$

$$= j\sqrt{\omega\mu\sigma}\sqrt{-j} = j\sqrt{\omega\mu\sigma}\left(\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}\right) \tag{2-5-7}$$

$$= (1 + j)\sqrt{\pi f\mu\sigma} \qquad [\omega = 2\pi f]$$

Hence

$$\alpha = \beta = \sqrt{\pi f\mu\sigma} \tag{2-5-8}$$

$\sigma = Sigma$ $\delta = Delta$

η= Eta

μ = Miu

Plane-Wave in Good Conductor

The exponential factor $e^{-\alpha z}$ of the traveling wave becomes $e^{-1} = 0.368$ when

$$z = \frac{1}{\sqrt{\pi f \mu \sigma}} \tag{2-5-9}$$

This distance is called the *skin depth* and is denoted by

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} = \frac{1}{\alpha} = \frac{1}{\beta} \tag{2-5-10}$$

Interestingly, at microwave frequencies the skin depth is extremely short and a piece of glass with an evaporated silver coating 5.40-μm thick is an excellent conductor at these frequencies.

The intrinsic impedance of a good conductor is given as

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \sqrt{\frac{j\omega\mu}{\sigma}} \qquad \text{for } \sigma \gg \omega\epsilon$$

$$\sqrt{j} = \left(\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}\right)$$

$$= \sqrt{\frac{\omega\mu}{\sigma}} \angle 45° = (1 + j)\sqrt{\frac{\omega\mu}{2\sigma}} \tag{2-5-11}$$

$$= (1 + j)\frac{1}{\sigma\delta} = (1 + j)R_s$$

in which $R_s = \sqrt{\omega\mu/(2\sigma)}$ is known as the *skin effect* and the magnitude of the conductor surface resistance.

Plane-Wave in Poor Conductor

Some conducting materials with low conductivity normally cannot be considered either good conductors or good dielectrics. Seawater is a good example. It has a conductivity of 4 mhos/m and a dielectric constant of 20. At some low frequencies the conduction current is greater than the displacement current, whereas at some high frequencies the reverse is true.

In general, the propagation constant and intrinsic impedance for a poor conductor are given by

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} \tag{2-5-14}$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} \tag{2-5-15}$$

Plane-Wave in Lossy Dielectric

All dielectric materials have some conductivity, but the conductivity is very small ($\sigma \ll \omega\epsilon$). When the conductivity cannot be neglected, the electric and magnetic fields in the dielectric are no longer in time phase. This fact can be seen from the intrinsic impedance of the dielectric as

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \sqrt{\frac{\mu}{\epsilon}}\left(1 - j\frac{\sigma}{\omega\epsilon}\right)^{-1/2} \tag{2-5-16}$$

The term $\sigma/(\omega\epsilon)$ is referred to as the *loss tangent* and is defined by

$$\tan\theta = \frac{\sigma}{\omega\epsilon} \tag{2-5-17}$$

This relationship is the result of displacement current density leading conduction current density by 90°, just as the current flowing through a capacitor leads the current through a resistor in parallel with it by 90° in an ordinary electric circuit. This phase relationship is shown in Fig. 2-5-1.

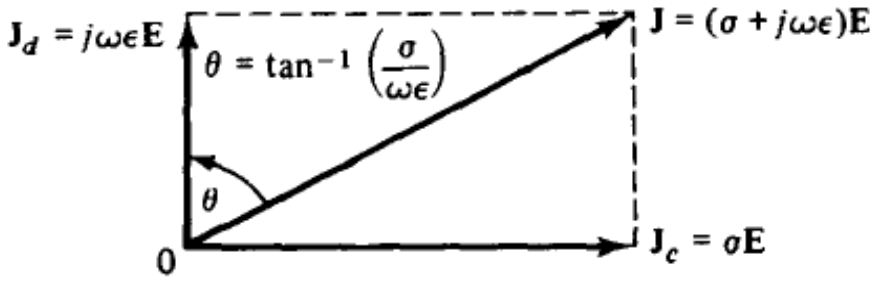


Figure 2-5-1 Loss tangent for lossy dielectric.

Plane-Wave in Lossy Dielectric

If the loss tangent is very small—that is, $\sigma/(\omega\epsilon) \ll 1$—the propagation constant and intrinsic impedance can be calculated approximately by a binomial expansion. Since

$$\gamma = j\omega\sqrt{\mu\epsilon}\sqrt{1 - j\frac{\sigma}{\omega\epsilon}} \tag{2-5-18}$$

then

$$\gamma = j\omega\sqrt{\mu\epsilon}\left[1 - j\frac{\sigma}{2\omega\epsilon} + \frac{1}{8}\left(\frac{\sigma}{\omega\epsilon}\right)^2 + \cdots\right] \tag{2-5-19}$$

Hence $$\alpha = j\omega\sqrt{\mu\varepsilon}(-j\frac{\sigma}{2\omega\varepsilon}) = \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}} \quad (2-5-20)$$

$$\beta = \omega\sqrt{\mu\varepsilon}[1+\frac{1}{8}(\frac{\sigma}{\omega\varepsilon})^2] = \omega\sqrt{\mu\varepsilon} \quad (2-5-21)$$

Plane-Wave in Lossy Dielectric

Similarly, $$\eta = \sqrt{\frac{\mu}{\varepsilon}}[1 + j\frac{\sigma}{2\omega\varepsilon} - \frac{3}{8}(\frac{\sigma}{\omega\varepsilon})^2 + \ldots] \quad (2-5-22)$$

or, $$\eta = \sqrt{\frac{\mu}{\varepsilon}}(1 + j\frac{\sigma}{2\omega\varepsilon}) \quad (2-5-23)$$

- Note that the attenuation constant for a given dielectric is not independent of frequency as shown in eq. (2-5-20). For high-frequency applications, the loss tangent varies slightly with frequencies.
- The reflectivity and transmittance of a lossy dielectric in vertical and horizontal polarization are commonly measured in terms of grazing angle.

Thank You